Ethan Deal

+1 (202) 560-2873 · ecd31@duke.edu · linkedin.com/in/ethdeal · ethdeal.github.io
Full-Stack Software Engineer | AI Platforms, MCP, React, TypeScript, Python, GCP

EDUCATION

Duke University & Duke Kunshan University | Dual Degree **May 2026 | Kunshan 昆山, CN**
B.S. Computation and Design, Computer Science

- GPA: 3.6/4.0; Awarded Merit Scholarship (60% Tuition)

EXPERIENCE

AllEmber Systems | Founding Full-Stack Engineer **Feb 2026 - Present | Remote**

- Co-founded an AI consultancy and provider-agnostic platform connecting company knowledge and workflows to AI applications including Claude and Microsoft Copilot, supporting **50+ users** across multiple organizations.
- Built and deployed a remote **MCP** service exposing custom tools and automated workflows to AI applications.
- Architected a multi-tenant platform where reusable tools support MCP interactions and scheduled automations; built provisioning tools for client onboarding, tool access, and configuration.
- Deployed services and background jobs on **GCP/Cloud Run**, with **CI/CD**, environment separation, and release-gated database migrations.
- Worked directly with client teams to map workflows, define requirements, and ship custom AI tools and automations.

BRIYA | Full-Stack Engineer **July - Aug 2025 | Washington, DC**
Previous role: IT Specialist **May - Aug 2024**

- Built and deployed a **MERN** stack room booking system supporting **5 campuses** and **100+ faculty**, achieving **zero downtime** since launch.
- Developed Node.js/MongoDB REST APIs for booking CRUD operations, recurring events, and conflict detection.
- Designed a React interface for desktop and mobile with multiple calendar views, drag-and-drop editing, and intuitive booking workflows.
- Deployed frontend and backend with GitHub Pages, Linode, SSL, Nginx reverse proxying, PM2, and DNS configuration.
- Previously enrolled and configured 1000+ Chromebooks, performed disk wipes, and supported device troubleshooting.

DUKE UNIVERSITY | Data Automation Engineer **May - July 2025 | Durham, NC**

- Built a Python/Pandas batch processing script to clean and segment **10,000+ mailing records**, exporting ready-to-use CSV reports.
- Developed a Python ETL tool to consolidate multi-event datasets into Excel reports for cross-event analysis.
- Automated Qualtrics workflows to centralize large-scale survey responses into a standardized spreadsheet.

PROJECTS

Research Automation Agent | AllEmber Systems **Aug 2026 - Present**

- Built an autonomous research agent for a client that monitors a shared inbox, detects new leads, and handles **5+ opportunities/week**.
- Designed the agent to break each lead into focused research tasks covering RFP context, organization identity, staffing, office footprint, financials, Form 990s, and relevant prior client work from internal knowledge sources.
- Synthesized findings into cited research briefs and recommendations, returning them directly in the original email thread.

Enterprise Data Ingestion & Retrieval | AllEmber Systems **May - Aug 2026**

- Built a configurable ingestion pipeline for diverse enterprise knowledge sources, continuously syncing **10,000+ documents**.
- Separated discovery and processing into parallel **Cloud Run Jobs** to efficiently sync large repositories without unnecessary reprocessing.
- Implemented permission-aware hybrid retrieval in Qdrant with Cohere reranking, enforcing source permissions during search.
- Built a self-service admin console for clients to authenticate, monitor, and manage enterprise data connections.

RAG Fact-Checking Extension **July 2025 - March 2026**

- Designed a Chrome extension in React with a Node.js/Express backend to analyze highlighted claims in under **5 seconds**.
- Architected a **RAG** pipeline using **LangChain**, ChromaDB, and a locally hosted MiniLM embedding model.
- Benchmarked single-step agentic vs. deterministic two-step retrieval architectures, evaluating latency, output control, context leakage, and susceptibility to injected misinformation.

SKILLS

Languages: Python, TypeScript, SQL, HTML/CSS, Java
Full-Stack: React, Next.js | Node.js, Express, FastAPI | PostgreSQL, MongoDB, MySQL
AI & Retrieval: MCP, LangGraph, Qdrant, Cohere, ChromaDB, LiteLLM
Cloud & Infrastructure: GCP, Docker, Supabase, Git, CI/CD

ADDITIONAL INFORMATION

Languages: English (native), Mandarin (intermediate), Spanish (beginner)
Interests: Ultimate frisbee, digital painting, travel photography, piano, guitar, soccer